

Joe Zhang 张洲

Male | Work Experience: 15 years at a global Top 1 MNC in pneumatics automation | System Analyst | Expected Salary: HKD \$40,000 per month

Proficient in English (listening, speaking, reading, writing), already obtained Hong Kong Quality Migrant Admission Scheme (QMAS) visa.

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Project Achievements

20+ Software Copyright Certificates | <https://github.com/0kinqoo/projects>

Education Background

- University of Technology Sydney, Australia (QS 2026 #96)
Master's Degree, Information Technology
- Xi'an University of Architecture and Technology, China
Bachelor's Degree, Information and Computing Science

Technical Skills Overview

Frontend Development

Proficient in Vue3 ecosystem (Vue3/Composition API/Vue Router 4/Pinia), compatible Vue2 development; skilled in using Element Plus/Ant Design Vue component libraries; proficient in Vite build optimization, Webpack performance tuning; familiar with JS/ES6+ advanced syntax (modularity, asynchronous programming, Proxy), TypeScript type system.

Backend Development

Proficient in .NET C# (ASP.NET Core 6.0+/EF Core 6.0+), full-stack development of .NET Core microservices architecture; familiar with PHP (Laravel/Swoole); skilled in RESTful API design, interface authentication (JWT/OAuth2), distributed system communication (gRPC/HTTP).

Database & Storage

Proficient in MSSQL (index optimization, transaction isolation, stored procedures), MySQL (sharding, master-slave replication, slow query analysis); familiar with Redis cache design (distributed locks, hot data preloading), Elasticsearch full-text search.

AI & Big Data

Deep practical experience with LangChain framework (Agent/Retrieval Chain/Tool Integration), full-process RAG (Retrieval Augmented Generation) development; proficient in vector database (Chroma/FAISS) applications, LLM (Large Language Model) API calls and fine-tuning; familiar with NLP basics (text cleaning, entity recognition).

Virtualization & Containerization & Tools

Proficient in containerization technologies, particularly Docker for efficient application deployment, environment isolation, and resource optimization. Familiar with virtualization concepts for system support and infrastructure management. Jenkins continuous integration; proficient in Postman API testing, Fiddler packet analysis; skilled in using Git (branch management/Rebase), familiar with Agile development processes (Jira/Confluence).

Power BI / Tableau Experience

Proficient in integrating various data sources (SQL Server, MySQL, APIs) into Power BI and Tableau, and transforming data using Power Query, DAX, and Tableau Prep. Experienced in building interactive, dynamic dashboards with advanced visualizations (e.g., maps, heat maps, scatter plots) and custom calculations (calculated fields, LOD, table calculations). Skilled in setting up Row-level Security (RLS) and publishing reports to Power BI Service or Tableau Server for collaboration and sharing. Familiar with embedding reports into web applications and optimizing performance for faster data access and real-time insights.

Work Experience

SMC (China) Co., Ltd. | Senior System Analyst (Feb. 2018 - Present)

*SMC is a global TOP 1 MNC in pneumatics automation.

SMC (China) Co., Ltd. | Software Development (Jun. 2011 - Jan. 2016)

Core Project Experience

AI-Powered Smart Customer Service System (.NET+Vue+LangChain RAG) (May 2024 - Present)

Project Background: Developed an AI-powered smart customer service platform for SMC, leveraging large language models to address product inquiries and after-sales issues. The project aimed to overcome limitations of traditional rule-based matching (only 30% coverage) and slow response times for complex issues (average 15s).

Technology Stack: Backend API: .NET Core 8.0, Frontend Interaction: Vue3 + Pinia, RAG Workflow Orchestration: LangChain, Database: MySQL, Vector Storage: Chroma, Fine-tuned Model: LLaMA-7B

My Responsibilities:

- Led RAG Module Design and Implementation: Completed business document (FAQ/product manual) cleaning (Regex + NER), vectorization (Sentence-BERT), and storage (Chroma vector database). Integrated LangChain's Retrieval Chain to achieve a closed-loop process of 'user query → semantic retrieval → answer generation', improving the resolution rate for complex issues from 30% to 85%.
- Backend API Development: Designed high-concurrency interfaces (peak QPS 2000+) based on ASP.NET Core Web API. Optimized database queries using EF Core (composite indexing + cache penetration prevention), reducing average interface response time from 800ms to 200ms.
- Frontend Interaction Optimization: Developed multi-turn dialogue components using Vue3 SFC + TypeScript, supporting context memory (LocalStorage caching) and voice input (Web Speech API), which increased user satisfaction by 40%.

Achievements:

- Post-launch, the system handled over 5,000 daily inquiries.
- Awarded the company's 'Annual Technology Innovation Award'.

LMS Learning Management Platform (Jun. 2022 - Present)

Project Background: Built a comprehensive Learning Management System (LMS) for enterprises, addressing challenges of traditional offline training such as difficult organization, weak performance tracking, and dispersed learning resources. The system supports online course learning, educational resource management, learning progress tracking, and assessment, ensuring stable operation under high-concurrency scenarios.

Technology Stack: Backend: .NET Framework 4.8, Frontend Framework: Vue.js, Database: MSSQL, Hangfire for asynchronous processing, SCORM standard courseware, Video Slicing Technology (HLS/MPEG-DASH), Multi-tenancy/Domain Partitioning.

My Responsibilities:

- Led Backend Architecture and Development: Designed and developed core business APIs for the platform using .NET Framework 4.8, integrated Hangfire for asynchronous task processing, and developed interfaces for SCORM courseware engine integration and data interaction.
- Database Design and Optimization: Designed database models tailored to LMS business requirements, performing index optimization and query performance tuning for high-frequency read/write scenarios (e.g., learning records, quiz data).
- Video Processing Workflow Implementation: Established an automated workflow for video upload, transcoding, slicing, and distribution, ensuring efficient and stable video playback on web and mobile platforms.

- **Frontend Feature Collaboration:** Collaborated with frontend engineers to define API specifications and jointly developed and debugged core interactive features such as course playback, learning progress display, and online examinations.

Achievements:

- Successfully delivered an enterprise-grade LMS platform, supporting over 10,000 concurrent online learners.

Pneumatic Smart Factory IoT SCADA Platform (Apr. 2023 - Present)

Project Background: Developed an IoT SCADA platform for pneumatic equipment in a smart factory for SMC and its clients. The project aimed to address issues of opaque equipment status, delayed fault response, and significant energy waste in traditional factories. By real-time monitoring of all pneumatic components, the platform enables fault prediction, remote diagnostics, and energy efficiency optimization, ultimately improving production efficiency and reducing operational costs.

Technology Stack: Frontend Framework: Vue.js (for building data visualization dashboards and monitoring panels), Industrial Protocol Gateway: Kepware KepserverEX, Communication Protocols: OPC UA (for standardized equipment data acquisition), MQTT (for efficient IoT data transmission)

My Responsibilities:

- **Led Overall Platform Solution Design:** Responsible for planning the comprehensive technical solution from device-level data acquisition to cloud application presentation, determining Kepserver as the core acquisition solution and a hybrid OPC UA/MQTT communication architecture.
- **Data Acquisition Gateway Configuration and Development:** Responsible for Kepserver configuration, tag definition, and debugging to ensure stable communication with various pneumatic component controllers on the factory floor; developed data forwarding modules to convert OPC UA data into MQTT messages.
- **Frontend Visualization Dashboard Development:** Developed monitoring dashboards using Vue.js and relevant visualization libraries (e.g., ECharts), implementing real-time equipment status updates, historical data chart tracing, and fault alarm pop-ups.
- **Fault Prediction and Energy Saving Algorithm Implementation:** Collaborated with algorithm engineers to integrate equipment fault models and energy consumption analysis algorithms into the platform's backend, developing corresponding API interfaces for frontend display of warnings and analysis results.

Achievements:

- Successfully launched the Smart Factory SCADA platform, enabling 24/7 real-time monitoring of over 5,000 pneumatic components across the entire factory.